

NAME SCHOOL TEACHER 

Pre-Junior Certificate Examination, 2016

Mathematics

Paper 2

Higher Level

Time: 2 hours, 30 minutes

300 marks

School stamp

Running total	
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For examiner			
Question	Mark	Question	Mark
1		11	
2		12	
3			
4			
5			
6			
7			
8			
9			
10		Total	

Grade

Instructions

There are 12 questions on this examination paper. Answer **all** questions.

Questions do not necessarily carry equal marks. To help you manage your time during this examination, a maximum time for each question is suggested. If you remain within these times you should have about 10 minutes left to review your work.

Write your answers in the spaces provided in this booklet. You may lose marks if you do not do so. There is space for extra work at the back of the booklet. You may also ask the superintendent for more paper. Label any extra work clearly with the question number and part.

The superintendent will give you a copy of the *Formulae and Tables* booklet. You must return it at the end of the examination. You are not allowed to bring your own copy into the examination.

You will lose marks if all necessary work is not clearly shown.

You may lose marks if the appropriate units of measurement are not included, where relevant.

You may lose marks if your answers are not given in simplest form, where relevant.

Write the make and model of your calculator(s) here:

Question 1

(Suggested maximum time: 15 minutes)

A standard DVD is in the shape of a circular disc of diameter 12 cm. It has a circular hole of diameter 1.8 cm in its centre, as shown.



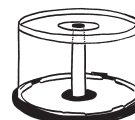
- (a) Find the **surface area** of one side of a DVD. Give your answer in terms of π .

[illegible]

- (b)** The case for a single standard DVD is 19 cm long, 13.5 cm wide and 1.4 cm high.
Find the **volume** of a single DVD case.

[illegible]

- (c) A DVD cake box holder is in the shape of a cylinder, as shown. It has an internal height of 6.1 cm. Given that the thickness of a DVD is 1.2 mm, find the **maximum** number of DVDs it can hold.

[illegible]

- (d) The external volume of the DVD cake box holder is 1384 cm^3 . It has an external height of 8.5 cm . Find the external **diameter** of the DVD cake box holder. Give your answer correct to one decimal place.

[illegible]

- (e) Express the volume saved by using a DVD cake box holder as a percentage of using single DVD cases to store the maximum number of DVDs that the DVD cake box holder can contain. Give your answer correct to two significant figures.

[illegible]

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Question 2

(Suggested maximum time: 20 minutes)

A simple random sample was taken from all the votes cast in last year's marriage referendum. The sample was taken from both urban and rural areas. The way in which votes were cast is shown in the two-way table below.



Vote	Urban Area	Rural Area	Total
Yes	602	Y	1014
No	X	388	786
Total	1000	800	1800

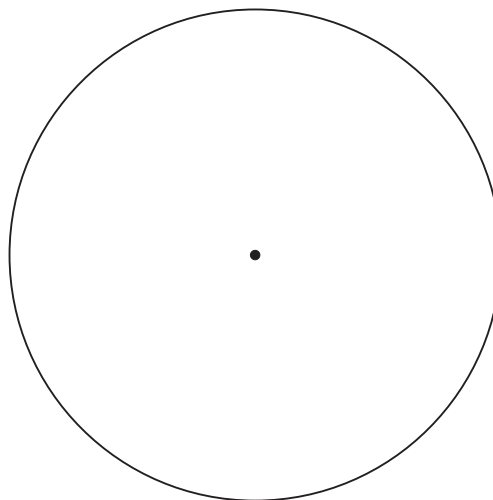
- (a)** Find the missing values for X and Y in the table above.

$X =$	$Y =$
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- (b)** Explain what is meant by the term **Simple Random Sample** in the context of this question.

[illegible]

- (c) Complete the pie chart to display the data from the sample above. Show all of your calculations clearly.

[illegible]

- [illegible]

- [illegible]

- [illegible]

- [illegible]

- Answer:**

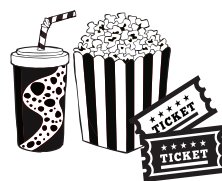
Reason:

Mathematics
Paper 2 – Higher Level

(Suggested maximum time: 5 minutes)

- [illegible]

- | Combi-Meal Menu | | |
|-----------------|------------|----------|
| Size | Soft Drink | Popcorn |
| Small | Orange | Plain |
| Medium | Cola | Salted |
| Large | Lemonade | Buttered |
| | Ginger Ale | Caramel |
| | Cream Soda | |

[illegible]

-
- $\frac{2}{9}$ **Medium** $\frac{1}{3}$ **Cola**
- $\frac{1}{2}$ **Plain Popcorn**
 - $\frac{1}{12}$ **Salted Popcorn**
 - $\frac{1}{4}$ **Buttered Popcorn**
 - $\frac{1}{6}$ **Caramel Popcorn**

- [illegible]

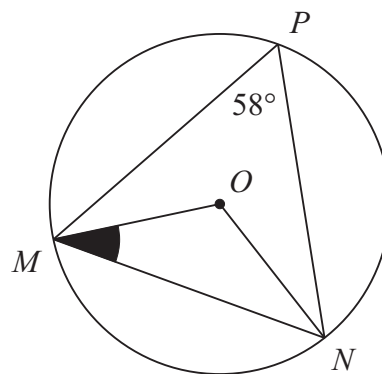
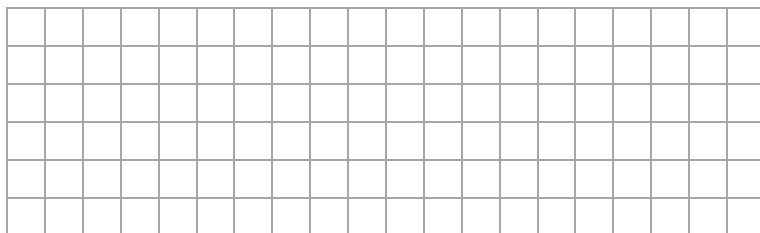
- [illegible]

Question 4

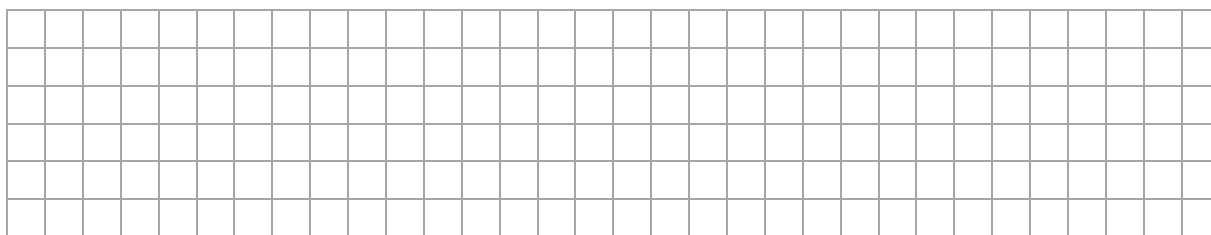
(Suggested maximum time: 10 minutes)

- (a) M, N and P are points on a circle with centre O .
 $|\angle NPM| = 58^\circ$, as shown.

- (i) Find $|\angle OMN|$.

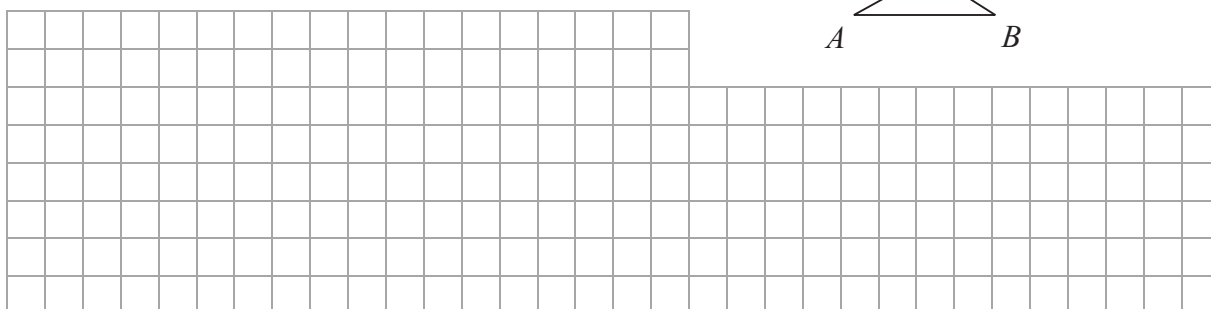
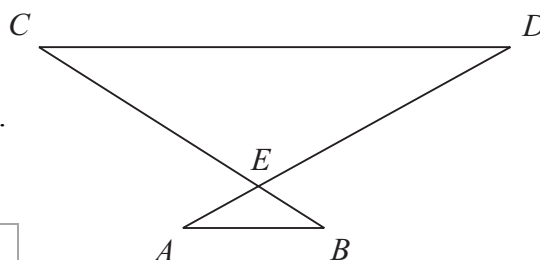


- (ii) Given that $|\angle PMO| = 29^\circ$, show that the triangle PMN is **isosceles**.

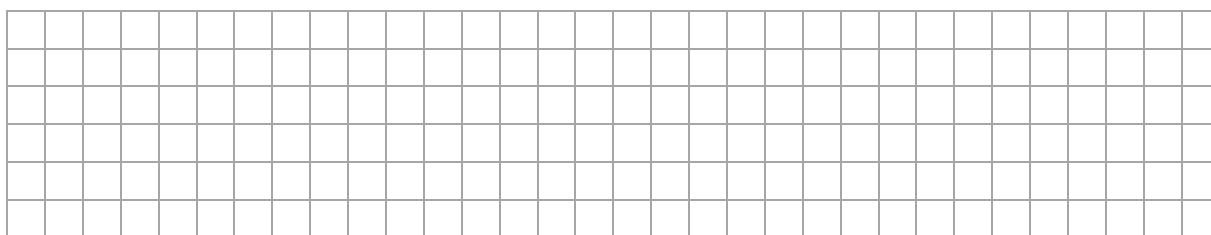


- (b) In the diagram, $[AB]$ is parallel to $[CD]$.
 $[AD]$ and $[CB]$ intersect at the point E .

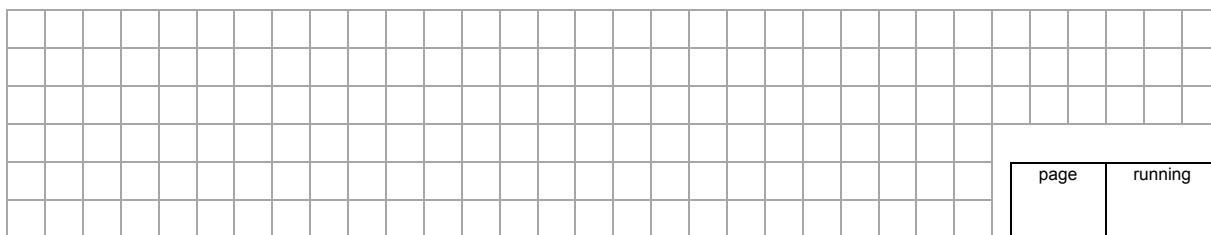
- (i) Prove that triangles ABE and CED are similar.
 Give a reason for each of the statements that you make in your proof.



- (ii) Given that $|\angle BEA| = 119^\circ$ and $|\angle DCE| = 32^\circ$, find $|\angle EAB|$.



- (iii) $|AE| = 16.5$, $|ED| = 55$ and $|CD| = 90$.
 Find $|AB|$.



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Question 5

(Suggested maximum time: 10 minutes)

Prove that, in a right-angled triangle, the square of the hypotenuse is equal to the sum of the squares of the other two sides.

Diagram:

<i>Given:</i>	
<i>To Prove:</i>	
<i>Construction:</i>	
<i>Proof:</i>	

Question 6

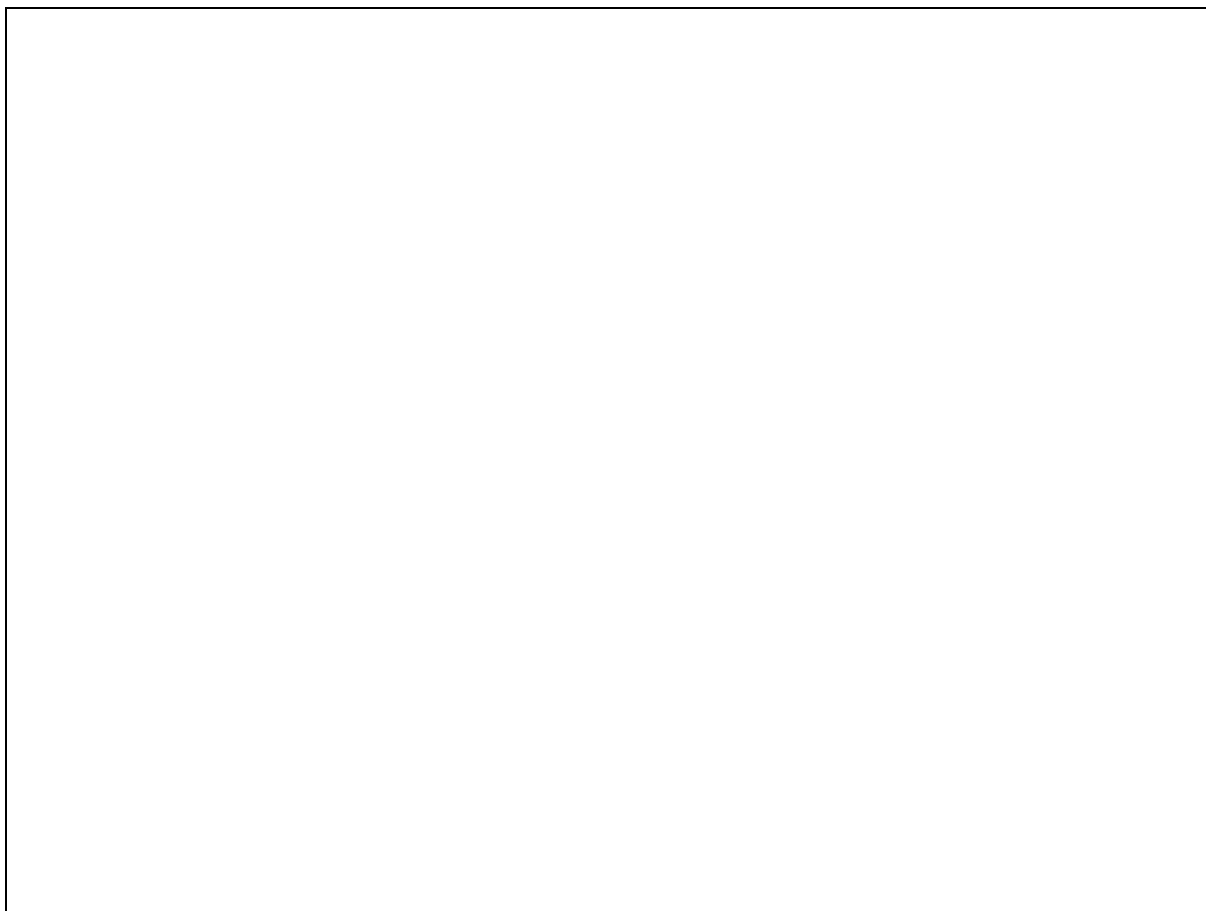
(Suggested maximum time: 10 minutes)

- (a) Construct a right-angled triangle ABC , where:

$$|\angle BAC| = 36^\circ$$

$$|AB| = 8 \text{ cm}$$

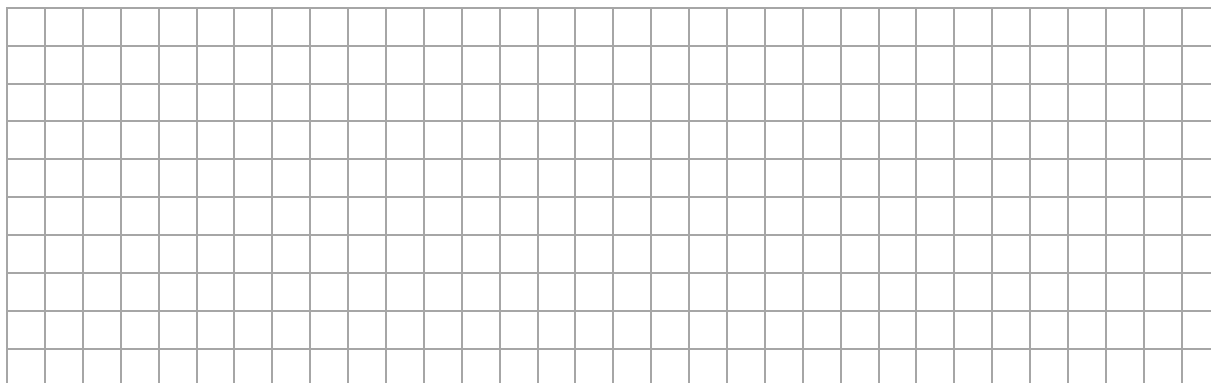
$$|\angle CBA| = 90^\circ.$$



- (b) On your diagram, measure the length of the side AC .
Give your answer in cm, correct to one decimal place.

$$|AC| = \underline{\hspace{2cm}}$$

- (c) Using trigonometry, verify your answer to part (b) above.

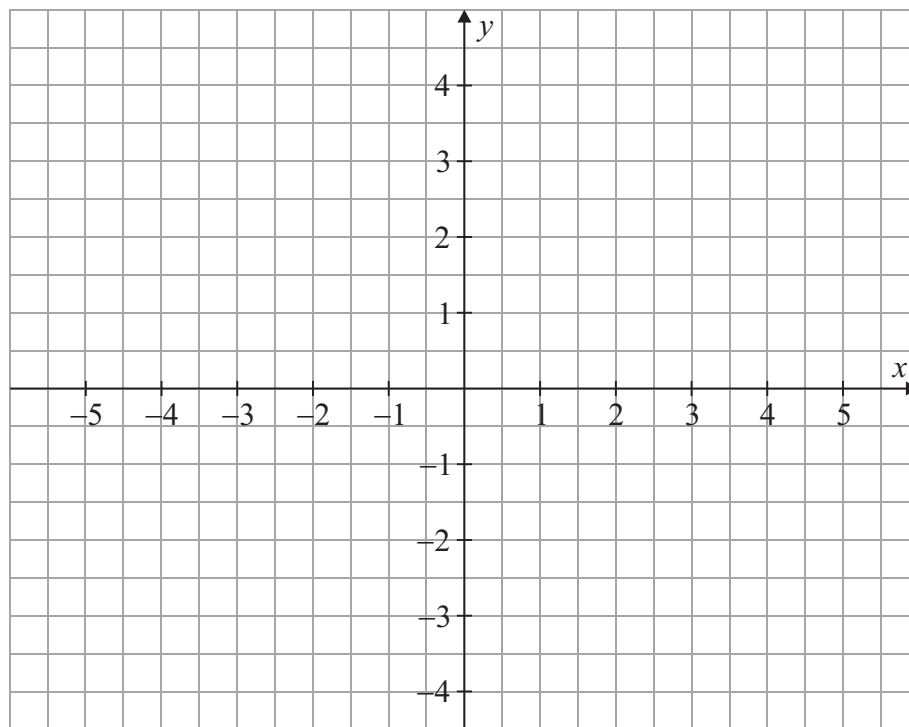


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Question 7

(Suggested maximum time: 15 minutes)

- (a)** Plot the points $A(-4, 1)$, $B(-4, 3)$ and $C(-1, 3)$ on the co-ordinate plane below. Join the points to form a triangle.



- (b)** What type of triangle does ABC represent? Give a reason for your answer.

[illegible]

- (c) Find the area of triangle ABC .

[illegible]

- (d)** Draw in the image of triangle ABC under central symmetry in the origin on the co-ordinate plane above.

- (e) State whether triangle ABC and its image under the transformation above are **similar** or **congruent**. Give a reason for your answer.

[illegible]

- (f) Is the area of the triangle ABC equal to the area of its image under the transformation above? Give a reason for your answer.

[illegible]

- (g) Find $|AC|$, giving your answer in surd form.

[illegible]

- (h)** Sean says that $\cos |\angle ACB|$ is equal to $\sin |\angle CAB|$. Is he correct? Give a reason for your answer.

[illegible]

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Question 8

(Suggested maximum time: 10 minutes)

The 'Monument of Light', alternatively known as the 'Spire', is a stainless steel, cone-shaped monument located in Dublin.

The Spire is the world's tallest sculpture at a height of 121.2 m. It was officially unveiled in 2003 at a total cost of €4 000 000 to Dublin City Council.



- (a) The circumference of the Spire at its base is 9.4 m.
Find the **radius** of its base, correct to one decimal place.

[illegible]

- (b)** Using the diagram, find l , the slant height of the Spire.
Give your answer in metres, correct to two decimal places.

[illegible]

- (c) Find the curved surface area of the Spire.
Give your answer in m^2 , correct to one decimal place.

[illegible]

- (d) The Spire cost the construction consortium that were awarded the contract €3 675 421 to manufacture and erect.
Find the percentage profit made by the consortium, correct to one decimal place.

[illegible]

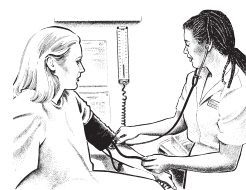
Question 9

(Suggested maximum time: 10 minutes)

The back-to-back stem-and-leaf plot below shows the systolic blood pressure of thirteen people before and two hours after taking a particular drug.

Before				After		
			10	9		
			11	1	6	
	8	6	12	5	7	
		4	13	1	7	
9	7	0	14	5	6	8
6	4	3	15	9		
	7	5	16	0		
		8	17			
		0	18	1		

Key: | 15 | 9 means 159 mm Hg



- (a) Find the **median** blood pressure both before and after taking the drug.

[illegible]

- (b)** Find the **range** of blood pressures both before and after taking the drug.

[illegible]

- (c) What **other** measure of variability (spread) could have been used when examining this data?

[illegible]

- (d) Compare the blood pressure results both before and after taking the drug. Refer to **at least one** measure of central tendency and **at least one** measure of variability (spread) in your answer.

A blank sheet of graph paper with a grid of squares. The grid consists of 20 columns and 10 rows of small squares. There are no margins or additional markings on the page.

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Question 10

(Suggested maximum time: 10 minutes)

In 1975, an American daredevil known as Evel Knievel attempted to jump over thirteen single-deck buses on his motorcycle in front of 90 000 people at Wembley Stadium in London.



To complete the jump, a launch ramp 3.5 m in height with a base 10.4 m in length was constructed, as shown below.



- (a) For Evel not to undershoot or overshoot the jump, he required the angle of elevation of the launch ramp to be between 15° and 20° . Would you conclude that the ramp met this criteria? Use calculations to justify your answer.

[illegible]

- (b)** Find the length of the ramp. Give your answer in metres, correct to two decimal places.

[illegible]

- (c) The width of each bus was 2.5 m. Evel estimated that he needed to travel at an average speed of 90 km/h in mid-air to complete the jump.
Find the minimum time he needed to travel in mid-air to complete the jump successfully.
Give your answer in seconds.

[illegible]

Question 11

(Suggested maximum time: 15 minutes)

The customers in a restaurant were surveyed to find out how long they each had to wait for a table on a particular night. The results are shown in the grouped frequency table below.

Time (minutes)	0 – 5	5 – 10	10 – 15	15 – 20	20 – 25
Frequency	20	11	9	7	3

Note: 10–15 means at least 10 minutes but less than 15 minutes, etc.

- (a)** What type of data was collected in the survey? Put a tick (✓) in the correct box below. Explain your answer.

Numerical
Discrete

Numerical
Continuous

Categorical
Nominal
□

Categorical
Ordinal
□

Explanation:

- (b)** Using mid-interval values, estimate the **mean** waiting time for a table on that particular night.

[illegible]

- (c) Display the above data on a histogram.

A full-page view of a blank sheet of graph paper. The grid consists of small, uniform squares formed by thin, light gray lines. There are no margins, text, or other markings on the page.

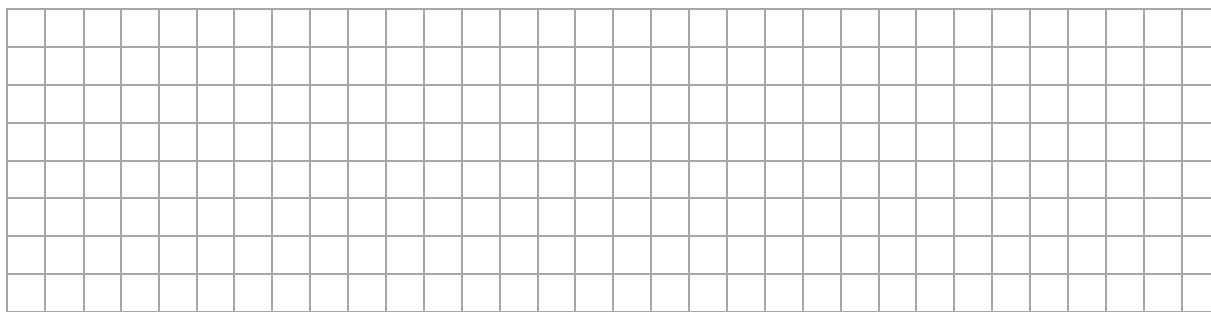
- (d)** What percentage of customers were waiting 15 minutes or longer for a table?

[illegible]

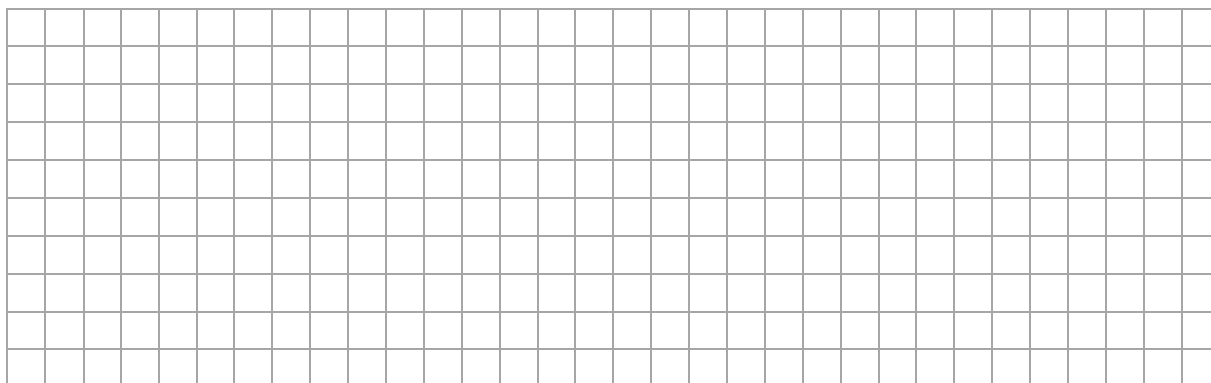
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Question 12**(Suggested maximum time: 10 minutes)**

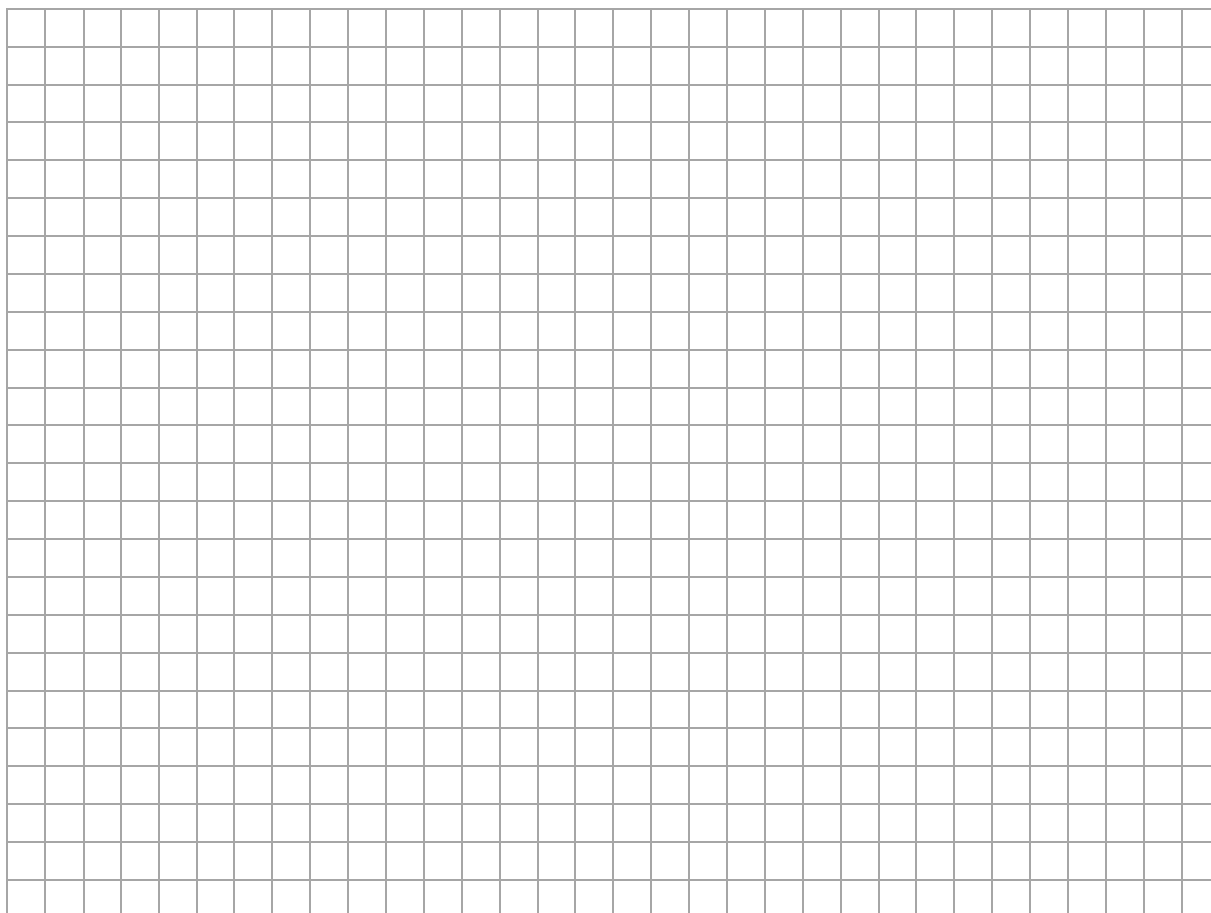
- (a) The equation of the line k is $2x + y - 1 = 0$.
Find the value of t such that $(2t, -3t)$ is a point on the line k .



- (b) The line l passes through the point $(3, 2)$ and is perpendicular to the line k .
Find the equation of the line l in the form $ax + by + c = 0$.



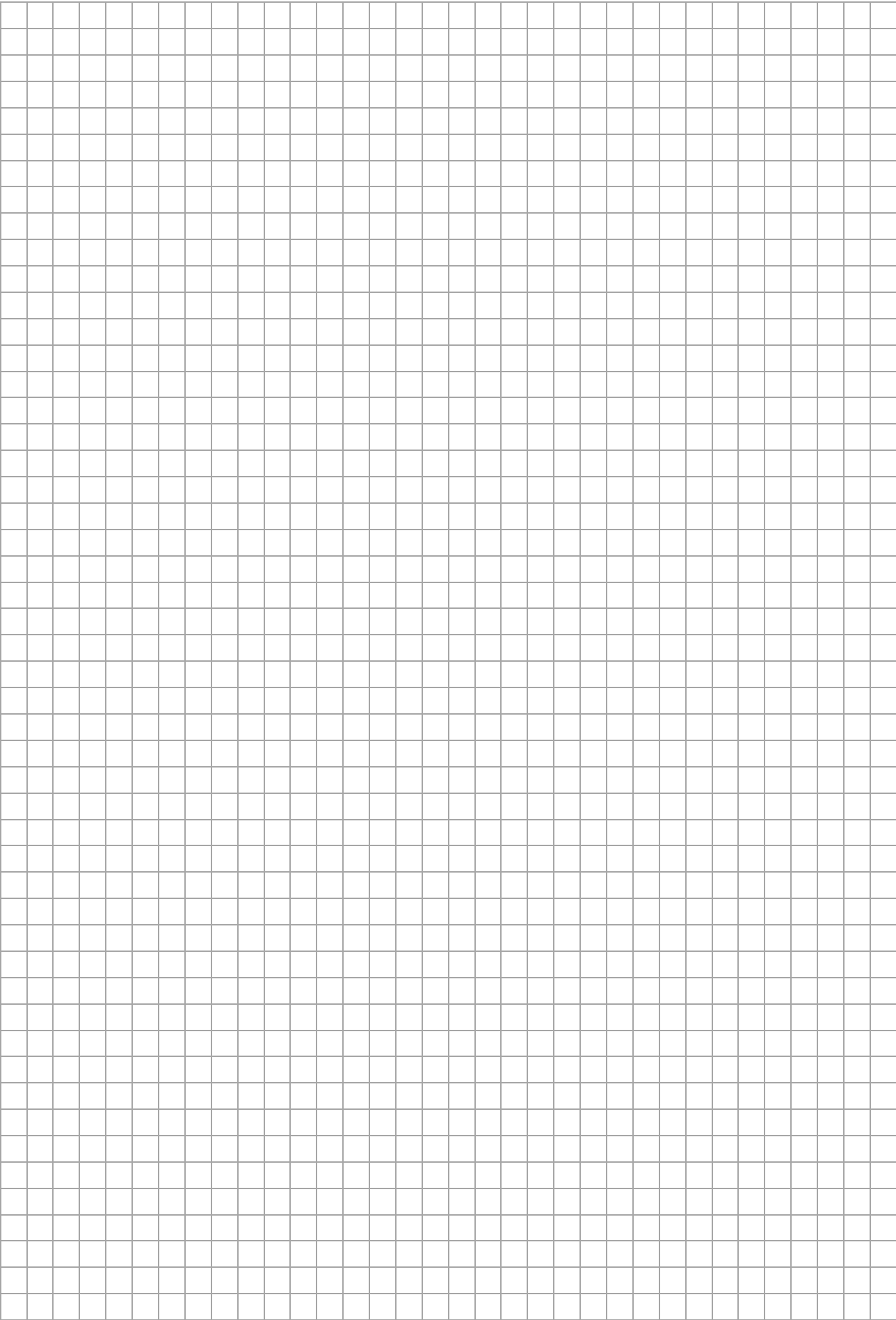
- (c) Find the co-ordinates of the point of intersection of lines l and k .



A full-page grid of graph paper. The grid consists of small squares. In the bottom right corner, there is a small table with two columns and two rows. The first row contains the words 'page' and 'running'. The second row is empty.

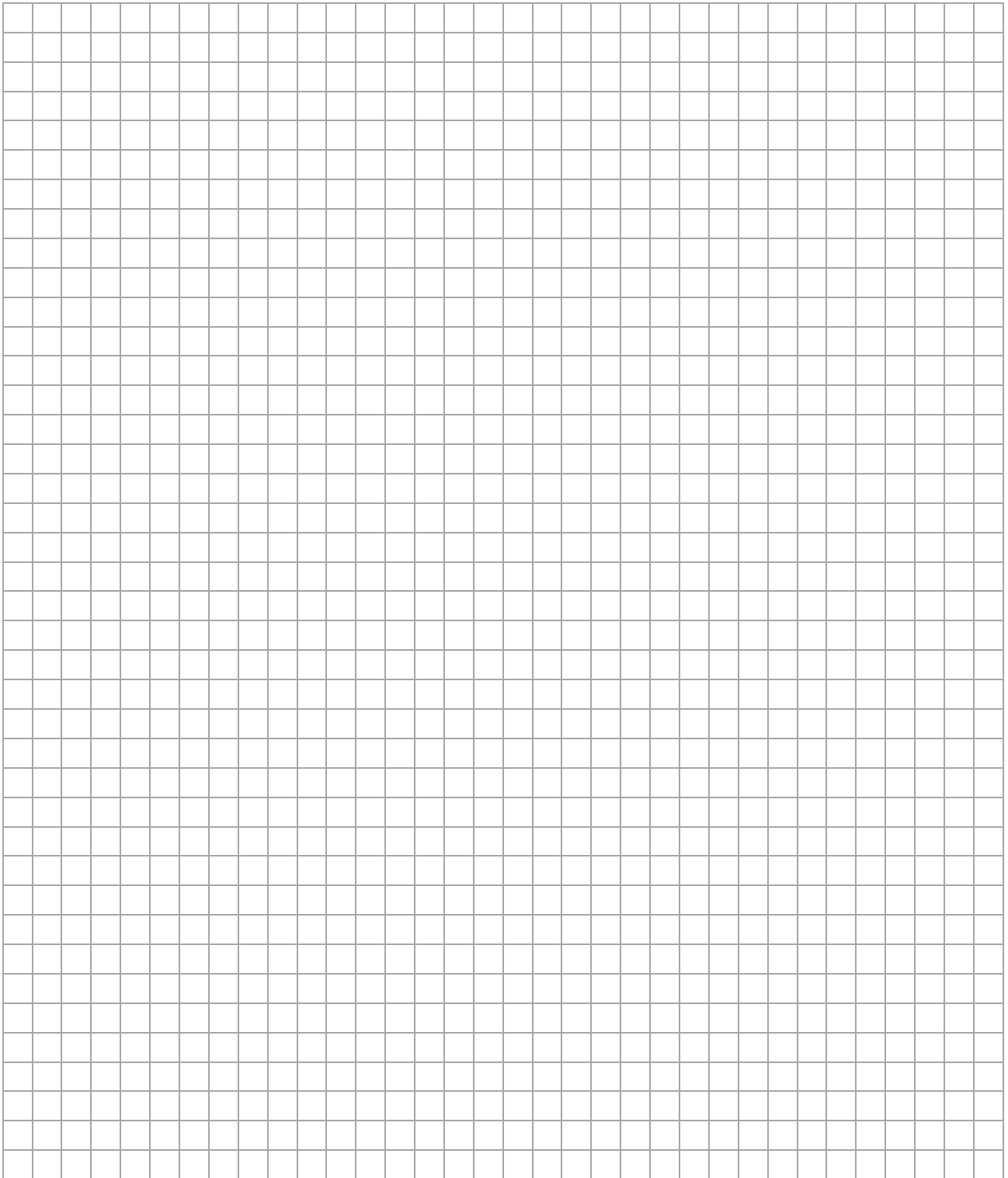
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You may use this page for extra work.



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Pre-Junior Certificate, 2016 – Higher Level

Mathematics – Paper 2

Time: 2 hours, 30 minutes

